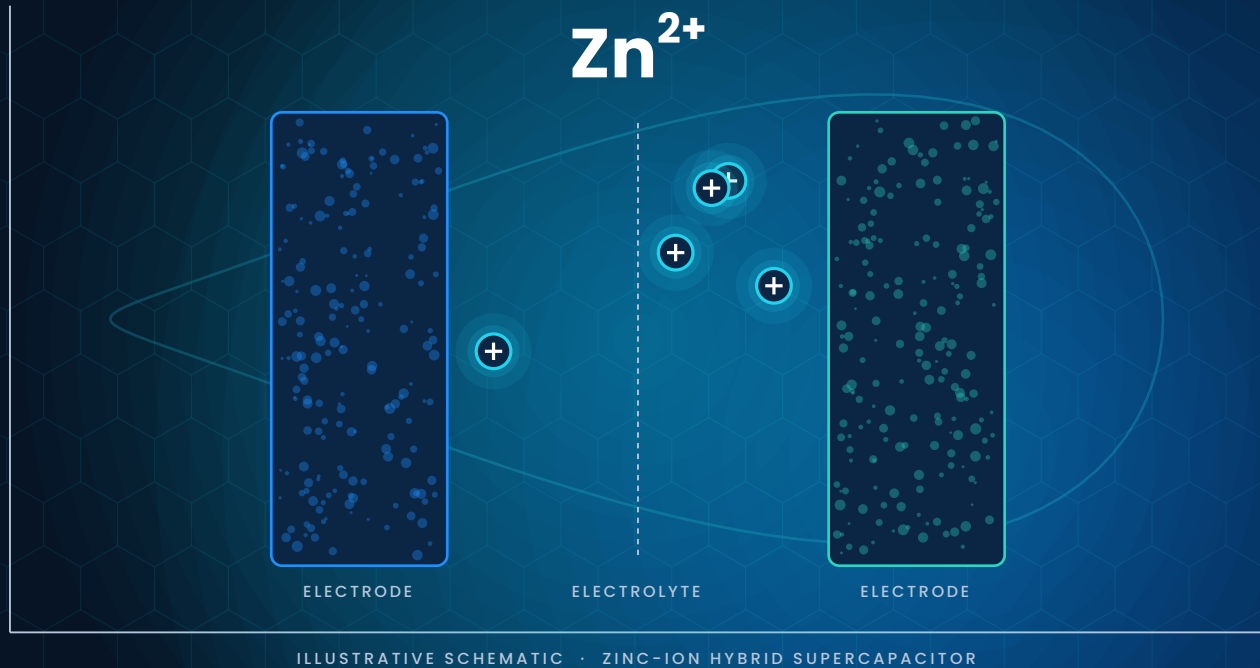




ONLINE WORKSHOP

ZINC-ION HYBRID SUPERCAPACITOR (ZHSC)

Fundamentals to Fabrication, Electrochemical Characterization
& Data Interpretation

Mr. Vineet Patel

Research Scholar, IIT Madras

Learn | Build | Innovate | Advance

17-18

OCTOBER 2026

6:00-8:00

PM IST

ONLINE

₹249

ABOUT THE WORKSHOP

Electrochemical energy storage begins with a simple question : how do we store charge, and how quickly can we deliver it? This two-day online workshop by Fluxiora Learning builds the answer step by step, starting with capacitors and batteries, comparing how each stores and releases energy, and arriving at supercapacitors and the Zinc-Ion Hybrid Supercapacitor (ZHSC).

From there, the programme moves from concepts to practice: electrode fabrication, electrochemical characterization by CV, GCD and EIS, three-electrode and two-electrode testing, and interpretation of data using Origin. It closes with applications, challenges and future scope.



WHY ZHSC?

Capacitors and batteries occupy different positions in energy storage, and the workshop begins by examining that contrast. Supercapacitors emerge from the need to bridge it, and the ZHSC is a hybrid configuration built around zinc-ion chemistry.

Understanding its components, terminology and characterization methods gives students a working foundation for research in energy storage and electrochemistry.



Illustrative schematic

THE LEARNING JOURNEY



WHAT YOU WILL LEARN

LEARNING OUTCOMES

By the end of the workshop, participants will be able to:

01 Fundamentals of Charge Storage

Explain the basics of capacitors, batteries and supercapacitors, and how they compare.



02 ZHSC Architecture

Describe the components, terminology and working concept of a Zinc-Ion Hybrid Supercapacitor.



03 Electrode Fabrication

Understand how electrodes are fabricated for supercapacitor devices.



04 CV, GCD & EIS

Understand what each electrochemical characterization technique measures and how it is used.



05 Testing Configurations

Understand three-electrode and two-electrode testing ($3E \rightarrow 2E$).



06 Data Interpretation with Origin

Explore how electrochemical data is plotted and interpreted using Origin software, including analysis of representative results.



07 Applications & Future Scope

Understand where supercapacitors are applied, and the challenges and future directions of the field.



TECHNICAL TOOLS / TECHNIQUES



CV



GCD



EIS



$3E \rightarrow 2E$



ORIGIN

Software is used for demonstration and interpretation; no software licence is included. Graphics are illustrative.

2-DAY TECHNICAL ROADMAP

DAY 1

17 OCT 2026
6:00–8:00 PM IST

FUNDAMENTALS & ZHSC

- 1 Basics of Capacitor
- 2 Basics of Battery
- 3 Capacitor vs. Battery
- 4 Need for Supercapacitors
- 5 Supercapacitor and its Types
- 6 Components of a Supercapacitor
- 7 Basic Terminology
- 8 Zinc-Ion Hybrid Supercapacitor (ZHSC)

DAY 2

18 OCT 2026
6:00–8:00 PM IST

FABRICATION, CHARACTERIZATION & DATA

- 1 Electrode Fabrication
- 2 Electrochemical Characterization Techniques
 - Cyclic Voltammetry (CV)
 - Galvanostatic Charge-Discharge (GCD)
 - Electrochemical Impedance Spectroscopy (EIS)
- 3 Three-Electrode & Two-Electrode Testing (3E → 2E)
- 4 Origin Software for Data Interpretation
- 5 Results / Representative Data
- 6 Applications of Supercapacitors
- 7 Challenges & Future Scope
- 8 Summary / Conclusion

CONCEPTS

PRACTICE

FROM ELECTROCHEMISTRY TO DATA

ELECTROCHEMICAL IMPEDANCE SPECTROSCOPY (EIS)

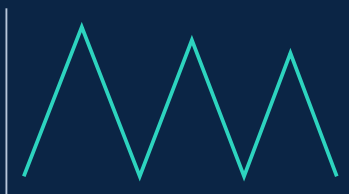
CYCLIC VOLTAMMETRY (CV)



Illustrative representation

The potential is swept back and forth while the current response is recorded. The shape of the resulting curve helps reveal how charge is stored in an electrode.

GALVANOSTATIC CHARGE-DISCHARGE (GCD)



Illustrative representation

The cell is charged and discharged at a constant current while the potential is tracked over time, showing how it behaves during charge/discharge cycles.



Illustrative representation

A small alternating signal is applied across a range of frequencies. The Nyquist plot separates resistive and capacitive behaviour.

FROM TESTING TO INTERPRETATION

3-ELECTRODE TESTING



2-ELECTRODE TESTING



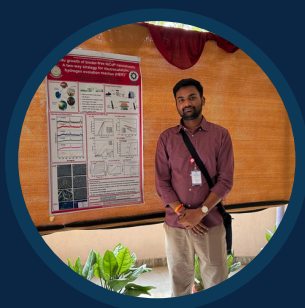
REPRESENTATIVE DATA



ORIGIN DATA
INTERPRETATION

Testing progresses from three-electrode measurements (3E) to two-electrode device-level testing (2E); the resulting data is then plotted and interpreted in Origin.

MEET THE SPEAKER



Mr. Vineet Kr. Patel

Research Scholar IIT Madras | INSPIRE Fellow
GATE & JAM Qualified

ABOUT THE SPEAKER

Mr. Vineet Patel is a Research Scholar at IIT Madras and an INSPIRE Fellow. He is GATE and JAM qualified and is engaged in research related to advanced materials and electrochemical energy-storage technologies. He will share research-oriented insights into Zinc-Ion Hybrid Supercapacitors (ZHSC), electrochemical characterization and data interpretation.

JOIN US

WHO SHOULD ATTEND?



Chemical Engineering



Materials Science



Physics



Chemistry



Research Scholars &
Academicians



Industrial R&D / Energy
Storage Professionals



Anyone interested in Energy Storage & Electrochemistry

WORKSHOP AT A GLANCE

DATE

17–18

October 2026

TIME

6:00–8:00

PM IST

MODE

Online

DURATION

2 Days

2 hours / day

FEE

₹249

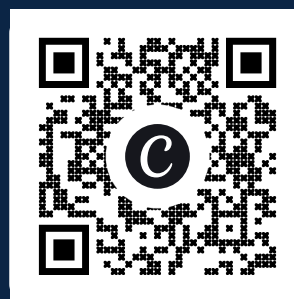
Registration

KEY TAKEAWAYS

- ✓ Strong conceptual understanding of supercapacitors and ZHSC
- ✓ Knowledge of electrode fabrication methods
- ✓ Understanding and interpretation of CV, GCD and EIS techniques
- ✓ Three-electrode to two-electrode (3E → 2E) testing concepts
- ✓ Data analysis and visualization using Origin
- ✓ Exposure to latest research trends, applications and future scope
- ✓ **Certificate of Participation (after successful completion)**

REGISTER NOW

SCAN TO REGISTER



₹249 Only

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